MIL-G-14934 (MU)
6 August 1968

MILITARY SPECIFICATION

GRENADE, HAND, FRAG, IMPACT, XM68

LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of grenade designated as Grenade, Hand, Frag, Impact, XM68.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein :

SPECIFICATIONS

MILITARY

MIL-A-2550	-	Ammunition and Special Weapons, General Specification for
MIL-I-45607	-	Inspection Equipment, Supply and Maintenance of

FSC: 1330

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STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-1168	-	Lot Numbering of Ammunition
MIL-STD-1235	-	Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

U.S. Army Munitions Command

9235493	-	Grenade, Hand, Frag, Impact, XM68

PUBLICATIONS

ARMY

ET-9235493	-	Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

49-CFR	171-178	-	Department of Transportation Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles

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(The Department of Transportation Regulations are now a part of the Code of Federal Regulations (1949 Edition and revisions) available from the Superintendent of Documents, U.S. Government Printing Office, Washington, D.C.20402. Orders for the above publications should cite, "49 CFR 171-178, (latest revision)".

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with the applicable drawings and specifications.

3.2 Parts.-The parts shall comply with all requirements specified on Drawing (dwg.) 9235493, all associated drawings, and with all requirements specified in applicable specifications.

3.3 Loading of high explosive.-

3.3.1 Each part assembly just before loading shall be inspected for presence of primer paint and no foreign matter. Parts assemblies with the interior wall incompletely coated or not dry shall be not be loaded.

3.3.2 Solid explosive shall not be present before loading or added during pouring or cooling.

3.3.3 The fuze seating surface shall be free of caked explosive. A chemical cleaner shall not be used.

3.3.4 Clean scrap produced in regular operations may be remelted and used provided it complies with the grade of explosive specified except for form and viscosity. Clean scrap is defined as scrap formed in kettles, pails, risers and loading machines used in pouring, easting and drilling operations.

3.4 Workmanship.-All parts and assemblies shall be fabricated, loaded and assembled in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, grease, rust, and other foreign matter. The cleaning method used shall not be injurious to any part, nor shall the parts be contaminated by the cleaning agents. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required markings shall be neat and sharply defined.

3.5 Submission of results of contractor-conducted examinations and tests.- The contractor shall forward copies of the results of any of his examinations or tests to the designated address as required by the contract.(See 6.4)

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4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his or any other facility suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 in order to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all the quality assurance provisions specified within this specification.
b. Number of units of product inspected.
c. Results obtained, by code number, for all inspections performed.
d. Drawing, specification number and date, together with an identification and date of changes.
e. Certificates of conformance on all material purchased by the contractor when such material is controlled by Government or commercial specifications referenced in any of the contractual documents.
f. Number of items in the lot.
g. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 First article inspection.- Not applicable

4.3 Inspection provisions.

4.3.1 Lot formation.-The term "Lot" as used throughout this specification refers to an inspection lot, which is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and

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inspected to determine conformance of the lot with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications, and same specification revisions. Changes to the process, specifications, or drawings, not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of the lot. Inspection lots shall comply with MIL-STD-105.

Unless otherwise approved by the Contracting Officer, the inspection lot size of end items deliverable under the contract shall be not less than the smallest weekly estimate of quantities contractually scheduled for production during the contract period, nor more than the largest monthly quantity contractually scheduled for delivery during the contract period. Inspection lots of components or subassemblies, other than the items of delivery, shall be homogeneous and of a size mutually convenient to both the contractor and the Government inspector. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each lot of grenades shall contain:

a. Metal parts and metal parts assemblies of one interfix lot number from one manufacturer.

b. Composition B of one interfix lot number from one manufacturer.

c. Fuzes from not more than two serial lots from one lot interfix number from one manufacturer. These lots shall be consecutively produced lots, or if the adjacent fuze serial lot is not available, the second fuze lot in the loaded grenade lot shall not vary from the first lot by more than three serial lot numbers.

4.3.2 Examination.-Sampling plans and procedures for the following classification of defects shall be in accordance with MIL-STD-105 (ABC-STD-105) except that, inspection for Critical defects shall be 100 percent. Contractor's sampling plans,if used, shall be approved by the Government, and shall provide as a minimum, the protection afforded the Government by the sampling plans in MIL-STD-105. Continuous sampling plans in accordance with MIL-STD-1235, may be used if approved by the procuring activity. Also, at the option of the procuring activity, AQL's and sampling plans may be applied to the individual characteristics listed, using an AQL of 0.40 percent for each Major defect and an AQL of 0.65 percent for each Minor defect, except where 100% inspection is specified. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

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4.3.2.1 Loading assembly (See Dwg. 8810742 covering a detail of Dwg. 9235493).-

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	AQL 0.40 Percent		
101.	Depth of fuze cavity, maximum (max.)	Gage	01001
102.	Diameter of fuze cavity, max.	Gage	01002
Minor	AQL 1.00		
201.	Explosive in threads	Visual	01003
202.	Explosive on fuze seating surface	Visual	01004

4.3.2.2 Grenade assembly (See Dwg. 9235493).-

Categories	Defects	Method of Inspection	Code No.
Critical			
1.	Safety pin not properly spread	Visual	02001
2.	Safety clip missing or improperly positioned	Visual	02002
Major	AQL 0.40 Percent		
101.	Pull ring missing	Visual	02003
Minor	AQL 2.50 Percent		
201.	Total height, maximum (max.)	Gage	02004
202.	Protective coating with bare spot	Visual	02005
203.	Washer missing	Visual	02006
204.	Marking misleading or unidentifiable	Visual	02007
205.	Evidence of poor workmanship (See 3.4)	Visual	02008

4.3.2.3 Fiber container, prior to sealing (See Dwg. 8836008 Sheet 1).-

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	AQL 0.40 Percent		
101.	Grenade inverted	Visual	03001
Minor	AQL 1.00 Percent		
201.	Fillers missing	Visual	03002
202.	Pad missing on top of grenade	Visual	03003

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4.3.2.4 Sealed fiber container (See Dwg. 8836008).-

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined.		
Major	AQL 0.65 Percent		
101.	Contents loose	Manual	04001
102.	Container damaged	Visual	04002
103.	Container improperly sealed	Visual	04003
Minor	AQL 0.65 Percent		
201.	Marking misleading or unidentifiable	Visual	04004

4.3.2.5 Box packing, prior to sealing (See Dwg. 8836008).-

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	AQL 0.40 Percent		
101.	Incorrect number of fiber containers	Visual	05001
102.	Container inverted	Visual	05002
Minor	AQL 0.65 Percent		
201.	Fillers missing	Visual	05003

4.3.2.6 Box, packing, sealed (See 8836007).-

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major	AQL 0.65 Percent		
101.	Contents loose	Manual	06001
102.	Strapping insecure	Manual-Visual	06002
103.	Box damaged	Visual	06003
104.	Box improperly sealed	Visual	06004
Minor	AQL 0.65 Percent		
201.	Marking misleading or unidentifiable	Visual	06005

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4.3.3 Testing.-

4.3.3.1 Specific gravity of the charge (See Dwg. 8810742), Major defect, Code No. 07001 .-Two (2) samples shall be selected from each four (4) hours of production and subjected to this test. If any sample fails to comply with the requirement specified on the applicable drawing, the shift's production represented by the sample shall be rejected. The test shall be performed as specified in 4.4.1.

4.3.4 Inspection equipment.-Equipment Tabulation (ET) number 9235493 and Equipment Lists (EL's) referenced thereon identify the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3.

4.3.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of this specification may be used by DOD activities for design, procurement, manufacture, testing, evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey, and whenever inspection equipment drawings are needed.

4.3.4.2 Supply and maintenance.-Supply and maintenance of the equipment listed on the EL's shall be in accordance with MIL-I-45607.

4.3.4.3 Government use of contractor's inspection and test equipment.-The contractor shall, when required by the Government, make available all inspection and test equipment necessary for determining conformance with the contract requirements. Personnel for operating the equipment, and verification of its accuracy shall be supplied by the contractor for the performance of examination or test by the Government.

4.4 Test methods and procedures.-

4.4.1 Specific gravity of the charge.-The specific gravity of Composition B shall be determined as follows:

Weigh the empty grenade assembly in grams and record as We. Fill the empty grenade assembly with water at 75°F ± 10°F, weigh in grams and record weight as Ww. Determine the volume by substracting We from Ww. Load the grenade with Composition B, drill the fuze well cavity, weigh in grams and record weight as Wl. Calculate density as follows:

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$$\text{Specific gravity} = \frac{Wl - We}{(Ww - We) - 8.2}$$

5. PREPARATION FOR DELIVERY

5.1 Preservation and Packaging.-

5.1 Level A.-Preservation and packaging shall be in accordance with Dwg. 8836007.

5.2 Packing.-

5.2.1 Level A.-The containers shall be packed in accordance with Dwg. 8836008.

5.3 Marking.-Marking shall be in accordance with Dwg.8796522 and Department of Transportation Regulations 49 CFR 171-178.

5.4 Shipping.-When shipments from more than one lot are shipped as a carload, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.

6.2 Inspection code numbers.-The five-digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.3 Inspection equipment.-The contractor shall design inspection equipment as required by the referenced Equipment Lists (EL) in accordance with the instructions of Paragraphs 6.3.1 through 6.3.6.

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6.3.1 Equipment Lists.-Equipment lists indicate the availability of inspection equipment designs by showing in the "Part number" column of the list of inspection equipment (Form SMUPA-1010) the numbers of drawings or Federl Stock numbers of existing equipment designs, or codes as indicated in Paragraph 6.3.2. Design action required of the contractor is described in Paragraphs 6.3.3 and 6.3.4. The contractor will be required to prepare detailed drawings in accordance with 6.3.4 for all the equipment coded as "Contractor Design" in the "Part Number" column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.6. Action required by the contractor with respect to the use of commercial equipment is detailed in 6.3.5.

6.3.2 Equipment list codes.-The inspection equipment as defined in 6.3.3, 6.3.4 and 6.3.5 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.3 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army designs fall into two basic classifications: mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an equipment list references mandatory Army designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, and with the approval of the Government, design alternate inspection equipment to facilitate his operations. Such contractor-prepared designs (See 6.3.4) or commercial equipment selections (See 6.3.5) must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.6.

6.3.4 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finishes, notes, and operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation will not be required. Unless otherwise specified, contractor

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designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to Specification MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high-contrast reproductions. Contractors shall submit one copy of final designs as a rolled set. Designs shall be submitted for approval as specified in 6.3.6.

6.3.5 Commercial equipment.-Commercial equipment is inspection equipment that has universal application for a specific function. It is comprised of items commonly used by industry and Government. Contractors, normally, are not required to furnish drawings of commercial inspection equipment, but a list of such equipment must be approved by the Government. When commercial inspection equipment requires the use of special fixturing, the design of this fixturing shall be considered as part of the total design, and 6.3.4 shall apply. In the latter case, only those portions of the commercial equipment need be detailed in the designs as are necessary to completely fabricate, calibrate, and operate the total item of inspection equipment. Lists and fixturing designs shall be submitted for approval as specified in 6.3.6.

6.3.6 Submission of contractor designs.-All submitted designs shall contain a reference to the applicable 5-digit code number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as Critical and Major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND13. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract. Submission shall be as directed by the Contracting Officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished, normally, within one month after receipt.

6.4 Submission of results of contractor-conducted examinations and tests.-Unless otherwise specified by the Contracting Officer, the contractor shall forward requested records of examinations or tests to Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND3.

Custodian:
Army-MU

Preparing activity:
Army-MU

Project Number: 1330-A115

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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